Minimal Aquaponics + Mariculture + Biofloc Product List

15 essential products/components for a basic monitoring and operating setup.

#	Product / Component	Use	Purpose
1	pH Sensor / pH Meter	Measures acidity/alkalinity	Core water-quality control; important for fish, shrimp and plants.
2	TDS / EC Sensor	Measures dissolved ionic content/conductivity	Tracks dissolved salts/minerals and changes in water quality.
3	Turbidity Sensor	Measures water cloudiness/suspended particles	Helps monitor solids/floc and filtration performance.
4	Dissolved Oxygen (DO) Sensor	Measures oxygen in water	Critical for fish/shrimp, plants and biofloc bacteria.
5	Water Temperature Sensor (DS18B20)	Measures water temperature	Tracks thermal conditions for culture and microbial activity.
6	Ammonia / TAN Test Kit	Checks ammonia level	Detects toxic nitrogen buildup before it harms stock.
7	Nitrite Test Kit	Checks nitrite	Monitors the ammonia-to-nitrate biological conversion process.
8	Salinity Sensor / Refractometer	Measures salinity	Essential for mariculture/brackish systems and species management.
9	Water Pump	Moves water between tanks/filters/grow beds	Provides circulation through the aquaponics loop.
10	Air Pump + Air Stones/Diffusers	Adds oxygen and mixes water	Maintains aeration and biofloc suspension.
11	Mechanical Filter / Settling Filter	Removes excess solids	Controls suspended waste before water reaches plant/biofilter stages.
12	Biofilter Media	Provides surface for beneficial bacteria	Supports nitrification and biological filtration.
13	Grow Bed / NFT Channels	Holds plants	Uses nutrient-rich aquaculture water for plant production.
14	Biofloc Carbon Source + Probiotic/Bacteria Starter	Supports microbial floc formation	Helps bacteria assimilate nitrogen and establish the biofloc system.
15	Water Test Kit / Alkalinity Test Kit	Checks alkalinity and supporting parameters	Helps maintain buffering capacity and stable pH.